How Plants Make Food, Drink Water and Help Make Rain

What Plants Need

Plants are living things, just like you. To stay alive and grow, a plant needs light, water, air (carbon dioxide), and minerals from the soil. A plant also needs space to grow. Animals have to go and find their food, but green plants are different: they make their own food. They do it inside their leaves using a process called photosynthesis. The word photosynthesis means "putting together with light." If you put a plant in a dark cupboard, it has water and air but no light, so it cannot make food. It slowly turns pale and weak.

Photosynthesis: The Leaf Kitchen

You can think of a leaf as a tiny kitchen. To make food, a plant needs three ingredients: water, carbon dioxide and light energy. Water comes up from the roots. Carbon dioxide is a gas that comes in from the air. Light energy comes from the Sun. Inside the leaf, the plant uses the light energy to join water and carbon dioxide together. This makes a sugar called glucose. Glucose is the plant's food. Oxygen is made too, and the plant lets most of it out into the air. Scientists write this as a word equation: carbon dioxide + water → glucose + oxygen. The arrow means "turns into", and the light energy powers the change. The plant uses glucose for energy and to build new leaves, stems, roots, flowers and fruit. It can store extra sugar as starch for later.

Chlorophyll and Chloroplasts

Leaves are green because they are full of chlorophyll. Chlorophyll is a green pigment that traps the energy in sunlight. It is found inside tiny parts of the leaf cells called chloroplasts. You can think of chloroplasts as the little solar panels of the leaf. Each leaf cell can have many chloroplasts. Leaves are usually wide and flat, so they can catch as much light as possible. That is why many plants turn their leaves to face the light.

Leaves, Stomata and Guard Cells

A leaf is not sealed shut. On its surface, mostly on the underside, there are thousands of tiny holes called stomata. One hole is called a stoma. Each stoma is like a tiny mouth with two doors. The doors are two bean-shaped cells called guard cells. Guard cells open the stoma and close it. When the stomata are open, carbon dioxide can come into the leaf, and oxygen and water vapour can go out. Stomata usually open in the daytime, when there is light for photosynthesis. They often close at night, and they close when the plant is short of water, to save water.

Roots, Xylem and Phloem

Roots hold the plant firmly in the ground and take in water and minerals from the soil. Roots are covered in tiny root hairs. Root hairs are very thin, so lots of them give the root a huge surface for soaking up water. Water travels up the plant through tubes called xylem. Xylem tubes run from the roots, up the stem and into every leaf, like a bundle of very thin straws. A second set of tubes, called phloem, carries sugar made in the leaves to all the other parts of the plant. Phloem takes sugar down to the roots, up to new buds, and into fruits and seeds. A simple way to remember it: xylem carries water up, phloem carries food around.

Transpiration: Plants Breathe Out Water

Most of the water a plant takes in does not stay inside it. When the stomata are open, water evaporates from inside the leaf and escapes into the air as water vapour. This loss of water from the leaves is called transpiration. Transpiration helps the plant in two big ways. First, as water leaves the top of the plant, more water is pulled up through the xylem to take its place. This is called the transpiration pull. It works a bit like sipping a drink through a straw: as water leaves the top, the whole line of water moves up behind it. This pull also carries minerals up from the roots. Second, evaporation cools the leaves, just like sweat cooling your skin on a hot day.

What Changes Transpiration

Plants do not transpire at the same speed all the time. Heat makes transpiration faster, because warm water evaporates more quickly. Wind makes transpiration faster, because it blows the water vapour away from the leaf. Humid air makes transpiration slower, because the air is already full of water vapour. Light makes transpiration faster, because the stomata open in the light. In the dark the stomata close, so transpiration slows right down. When soil is very dry, a plant closes its stomata to save water, and it may start to wilt.

Evapotranspiration and the Water Cycle

Water does not only leave the land through plants. Water also evaporates from soil, ponds, rivers, lakes and wet leaves. Scientists add the two together. Evapotranspiration is evaporation from soil and water plus transpiration from plants. In a big forest, plants send a huge amount of water vapour into the air every day. One large tree can release hundreds of litres of water on a hot summer day. This water vapour rises with warm air. High up, where the air is cooler, the water vapour cools and condenses into tiny droplets. This is called condensation. Billions of droplets together make clouds. When the droplets join and become heavy, they fall as rain. This is called precipitation. The rain soaks into the soil, the roots drink it up, and the water cycle starts again.

Why Plants and Forests Matter

Photosynthesis is the start of almost every food chain. A rabbit eats grass, and a fox eats the rabbit, so the energy in the fox's food first came from sunlight caught by a plant. Plants also give us oxygen to breathe and take carbon dioxide out of the air. Forests are like giant water pumps. The water vapour they breathe out helps to make clouds, and those clouds bring rain to the land, sometimes far away. When large areas of forest are cut down, this is called deforestation. With fewer trees there is less transpiration, so less water vapour goes into the air, and some places can get less rain. Planting trees and caring for forests helps keep the water cycle turning and keeps the land green.